A LOCAL RING IS GEOMETRICALLY UNIBRANCH EXACTLY WHEN ITS STRICT HENSELIZATION HAS A UNIQUE MINIMAL PRIME

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Allowed standard prerequisites (editorial).

- Flatness of the strict henselization makes the contraction of the unique minimal prime the unique minimal prime below, by going down. used at proof lines 31469-31472. Produces the unique minimal prime of the ring and the identification of the minimal prime of the strict henselization with its extension.
- The quotient of the strict henselization by a prime is the strict henselization of the quotient and is reduced, so the minimal prime extends. invoked at proof lines 31472-31476. Identifies the minimal prime of the strict henselization as the extension of the prime below.
- A finite local subalgebra of the integral closure obtained by adjoining finitely many elements keeps the relevant defect, and tensoring with the strict henselization turns it into a product of at least two local rings. used at proof lines 31482-31505. Produces the finite subalgebra B that leads to the contradiction with being a subring of a domain.
- For a strictly henselian local ring, a finite algebra whose residue ring has two maximal ideals or a non purely inseparable residue field splits as a product. invoked at proof lines 31504-31507. Gives the splitting of the tensor product which contradicts its lying inside a domain.
- Localizations of etale algebras preserve integrality, normality and the residue field conditions used in the converse. invoked at proof lines 31515-31520. Lets the converse pass from the integral closure of the ring to the tensor product with the localization of an etale algebra and conclude that the quotient is a domain.

Source reference. Original lemma 0C25: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31339–31339. The opening sentence notes the statement also follows from it, but the author gives the direct proof immediately; every step of that direct proof, including the contradiction with the splitting, is inside the counted span.

Source reference. Original lemma 0BQ0: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31249–31249. Cited at proof lines 31467-31469 only to say the direct proof is almost the same as the earlier one; the proof itself is written out in full inside the counted span.

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Source reference. Original lemma 06DH: sources/stacks-more-algebra/pinned-original-source.tex, source lines 11067–11067. Used at proof lines 31473-31475 to conclude that the quotient of the strict henselization by the prime below is reduced, which identifies the minimal prime above as the extension; the rest of the reduction is inside the counted span.

Source reference. Original lemma 00HS: sources/stacks-chow/tags.txt, source lines 1–1. Applied at proof lines 31469-31472 to see that the contraction of the minimal prime above is the unique minimal prime below; the use of that uniqueness is inside the counted span.

Source reference. Original lemma 05WS: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 31472-31474 to identify the quotient of the strict henselization by the prime with the strict henselization of the quotient; the conclusion drawn from that identification is inside the counted span.

Source reference. Original lemma 00GU: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 31480-31482 to see that every maximal ideal of the integral closure lies over the maximal ideal of the ring; the construction of the finite subalgebra and the contradiction are inside the counted span.

Source reference. Original lemma 06DD: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 31504-31507 to conclude that the tensor product with the strict henselization is a product of at least two local rings; producing the two maximal ideals of the residue tensor product is inside the counted span.

Source reference. Original lemma 092Y: sources/stacks-chow/tags.txt, source lines 1–1. Used in the converse at proof lines 31517-31519 to see that the tensor product with the integral closure stays local; the surrounding argument that it is a normal domain and that the quotient is a domain is inside the counted span.

Source reference. Original lemma 033C: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 31520-31523 to conclude that the tensor product is normal; the conclusion that it is a local normal domain and everything drawn from it is inside the counted span.

Lemma 0.2. *Let A be a local ring. Let A^{sh} be a strict henselization of A. The following are equivalent*

(1) *A is geometrically unibranch, and*
(2) *A^{sh} has a unique minimal prime.*

Proof. This follows from Lemma 0C25 but we will also give a direct proof; this direct proof is almost exactly the same as the direct proof of Lemma 0BQ0. Denote $\mathfrak{m}$ the maximal ideal of the ring A. Denote κ, κ^{sh} the residue field of A, A^{sh}.

Assume (2). Let $\mathfrak{p}^{sh}$ be the unique minimal prime of A^{sh}. The flatness of $A \to A^{sh}$ implies that $\mathfrak{p} = A \cap \mathfrak{p}^{sh}$ is the unique minimal prime of A (by going down, see Algebra, Lemma 00HS). Also, since $A^{sh}/\mathfrak{p}A^{sh} = (A/\mathfrak{p})^{sh}$ (see Algebra, Lemma 05WS) is reduced by Lemma 06DH we see that $\mathfrak{p}^{sh} = \mathfrak{p}A^{sh}$. Let A' be the integral closure of $A/\mathfrak{p}$ in its fraction field. We have to show that A' is local and that its residue field is purely inseparable over κ. Since $A \to A'$ is integral, every maximal ideal of A' lies over $\mathfrak{m}$ (by going up for integral ring maps, see Algebra, Lemma 00GU). If A' is not local, then we can find distinct maximal ideals $\mathfrak{m}_1$, $\mathfrak{m}_2$. Choosing elements $f_1, f_2 \in A'$ with $f_i \in \mathfrak{m}_i$, $f_i \notin \mathfrak{m}_{3-i}$ we find a finite subalgebra $B = A[f_1, f_2] \subset A'$ with distinct maximal ideals $B \cap \mathfrak{m}_i$, $i = 1, 2$. If A' is local with maximal ideal $\mathfrak{m}'$, but $A/\mathfrak{m} \subset A'/\mathfrak{m}'$ is not purely inseparable, then we can find $f \in A'$ whose image in $A'/\mathfrak{m}'$ generates a finite, not purely inseparable extension

of $A/\mathfrak{m}$ and we find a finite local subalgebra $B = A[f] \subset A'$ whose residue field is not a purely inseparable extension of $A/\mathfrak{m}$. Note that the inclusions

$$A/\mathfrak{p} \subset B \subset \kappa(\mathfrak{p})$$

give, on tensoring with the flat ring map $A \to A^{sh}$ the inclusions

$$A^{sh}/\mathfrak{p}^{sh} \subset B \otimes_A A^{sh} \subset \kappa(\mathfrak{p}) \otimes_A A^{sh} \subset \kappa(\mathfrak{p}^{sh})$$

the last inclusion because $\kappa(\mathfrak{p}) \otimes_A A^{sh} = \kappa(\mathfrak{p}) \otimes_{A/\mathfrak{p}} A^{sh}/\mathfrak{p}^{sh}$ is a localization of the domain $A^{sh}/\mathfrak{p}^{sh}$. Note that $B \otimes_A \kappa^{sh}$ has at least two maximal ideals because $B/\mathfrak{m}B$ either has two maximal ideals or one whose residue field is not purely inseparable over κ, and because κ^{sh} is separably algebraically closed. Hence, as A^{sh} is strictly henselian we see that $B \otimes_A A^{sh}$ is a product of ≥ 2 local rings, see Algebra, Lemma 06DD. But we've just seen that $B \otimes_A A^{sh}$ is a subring of a domain and we get a contradiction.

Assume (1). Let $\mathfrak{p} \subset A$ be the unique minimal prime and let A' be the integral closure of $A/\mathfrak{p}$ in its fraction field. Let $A \to B$ be a local map of local rings which is a localization of an étale A-algebra. In particular $\mathfrak{m}_B$ is the unique prime containing $\mathfrak{m}_A B$. Then $B' = A' \otimes_A B$ is integral over B and the assumption that $A \to A'$ is local with purely inseparable residue field extension implies that B' is local (Algebra, Lemma 092Y). On the other hand, $A' \to B'$ is the localization of an étale ring map, hence B' is normal, see Algebra, Lemma 033C. Thus B' is a (local) normal domain. Finally, we have

$$B/\mathfrak{p}B \subset B \otimes_A \kappa(\mathfrak{p}) = B' \otimes_{A'} (\text{fraction field of } A') \subset \text{fraction field of } B'$$

Hence $B/\mathfrak{p}B$ is a domain, which implies that B has a unique minimal prime (since by flatness of $A \to B$ these all have to lie over $\mathfrak{p}$). Since A^{sh} is a filtered colimit of the local rings B it follows that A^{sh} has a unique minimal prime. Namely, if $fg = 0$ in A^{sh} for some non-nilpotent elements f, g, then we can find a B as above containing both f and g which leads to a contradiction. $\square$